Wei-Chen Chen

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EXPERIENCE

Food and Drug Administration, USA.

- 08/2016–present, Mathematical Statistician, Center for Devices and Radiological Health.
- 08/2014–08/2016, ORISE Fellow, Center for Biologics Evaluation and Research.

Department of Ecology and Evolutionary Biology, University of Tennessee-Knoxville, USA.

- 09/2013–07/2014, Postdoctoral Associate.

Computer Science and Mathematics Division, Oak Ridge National Laboratory, USA.

- 08/2011–08/2013, Postdoctoral Research Associate.

Division of Biostatistics and Bioinformatics, National Health Research Institutes, Taiwan.

- 01/2001–08/2005, Research Assistant.

EDUCATION

- 08/2005–07/2011, Ph.D., Statistics, Iowa State University, USA.
- 09/1998–06/2000, M.S., Statistics, National Central University, Taiwan.
- 09/1994–06/1998, B.S., Mathematics, National Central University, Taiwan.

COMPUTER SKILLS

- **Statistical Software**: R, JAGS, WinBUGS.
- **Programming Languages**: C, Fortran, Perl.
- **Web Development Tools**: PHP, Html, CGI.
- **Database**: MySQL.
- **Operating Systems**: Linux/Unix, Solaris, MS-Windows.
- **Parallel Computing Libraries**: OpenMPI, MPICH, MS-MPI.
- **Network Related**: Apache, SSL, Samba, NFS, NIS, SSH, ZeroMQ.
- **Others**: TeX/LaTeX, Shell Script, vi, Word, Excel, PowerPoint.

PUBLICATIONS

Software Released on CRAN

- 11 pbdR Packages: **pbdRPC**, **remoter**, **pbdZMQ**, **pbdPROF** (GSoC sponsored (2013, 2014)), **pbdDEMO**, **pbdNCDF4**, **pbdDMAT**, **pbdBASE**, **pbdSLAP**, **pbdMPI**, and **pmclust**.
- 6 R Packages: **getPass**, **cubfits**, **QZ**, **EMCluster**, **phyclust**, and **MixSim**.

Refereed Papers

[13] Schmidt, D., **Chen, W.-C.**, Matheson, M., and Ostrouchov, G., "Programming with BIG Data in R: Scaling Analytics from One to Thousands of Nodes," *Big Data Research*, (accepted).

[12] Li, J.X., **Chen, W.-C.** and Scott, J.A. (2016), "Addressing Prior-data Conflict with Empirical Meta-Analytic-Predictive Priors in Clinical Studies with Historical Information," *Journal of Biopharmaceutical Statistics*, 16(6), 1056–1066.

[11] Schmidt, D., **Chen, W.-C.**, and Ostrouchov, G. (2016), "Introducing a New Client/Server Framework for Big Data Analytics with the R Language," *Proceedings of the XSEDE16 Conference on Diversity, Big Data, and Science at Scale*, 38:1-38:9.

[10] Gilchrist, M.A., **Chen, W.-C.**, Shah, P., Landerer, C.L., and Zaretzki, R. (2015), "Estimating Gene Expression and Codon-Specific Translational Efficiencies, Mutation Biases, and Selection Coefficients from Genomic Data Alone," *Genome Biology and Evolution*, **7**(6), 1559–1579.

[9] **Chen, W.-C.**, Ostrouchov, G., Pugmire, D., Prabhat, and Wehner, M. (2013), "A Parallel EM Algorithm for Model-Based Clustering Applied to the Exploration of Large Spatio-Temporal Data," *Technometrics*, **55**(4), 513–523.

[8] Melnykov, V., **Chen, W.-C.**, and Maitra, R. (2012), "MixSim: An R Package for Simulating Data to Study Performance of Finite Mixture Modeling and Clustering Algorithms," *Journal of Statistical Software*, **51**(12), 1–25.

[7] Schmidt, D., Ostrouchov, G., **Chen, W.-C.**, and Patel, P. (2012), "Tight Coupling of R and Distributed Linear Algebra for High-Level Programming with Big Data," *High Performance Computing, Networking, Storage and Analysis (SCC), 2012 SC Companion:*, 811-815.

[6] **Chen, W.-C.** and Maitra, R. (2011), "Model-Based Clustering of Regression Time Series Data via APECM – An AECM Algorithm Sung to an Even Faster Beat," *Statistical Analysis and Data Mining*, **4**(6), 567–578. (Student Paper Competition Winner).

[5] Carpenter, S., **Chen, W.-C.**, and Dorman, K.S. (2011), "Rev Variation during Persistent Lentivirus Infection," *Viruses*, **3**, 1–12.

[4] **Chen, W.-C.** and Maiti, T. (2007), "Analysis of cDNA Microarray Experiment with External Controls: A Mixed Model Approach," *Calcutta Statistical Association Bulletin*, **59**, 117–135 (by invitation).

[3] Wen, C.-C., Wu, Y.-J., Huang, Y.-H., **Chen, W.-C.**, Liu, S.-C., Jiang, S.-S., Juang, J.-L., Lin, C.-Y., Fang, W.-T., Hsiung, C.A., and Chang, I.-S. (2006), "A Bayes Regression Approach to Array-CGH Data," *Statistical Applications in Genetics and Molecular Biology*, **5**(1), Article 3.

[2] Fan, T.-H. and **Chen, W.-C.** (2005), "A Bayesian Change Points Analysis of the Seismic Activity in Northeastern Taiwan," *Journal of Statistical Computation and Simulation*, **75**(11), 857–868.

[1] Juang, J.-L., Chen, T.-C., Jiang, S.-S., Hsiung, C.A., **Chen, W.-C.**, Chen, G.-W., Lin, S.-M., Lin, J.-H., Chiu, S.-C., and Lai, Y.-K. (2004), "Coupling Multiplex RT-PCR to a Gene Chip Assay for Sensitive and Semiquantitative Detection of Severe Acute Respiratory Syndrome-Coronavirus," *Laboratory Investigation*, **84**(9), 1085–1091.

Conference Presentations

[7] **Chen, W.-C.** and Lu, N. (2017), "A Bayesian Approach Developed for Single Arm Studies using Performance Goals with Subgroups," Section on Medical Devices and Diagnostics, JSM2017, Baltimore,

MD, USA.

[6] **Chen, W.-C.**, Schmidt, D., and Ostrouchov, G. (2016), "Interactive Terabytes with pbdR," UseR! 2016, Stanford, CA, USA.

[5] **Chen, W.-C.**, Ostrouchov, G., Pugmire, D., Prabhat, and Wehner, M. (2012), "Model-Based Clustering Analysis of Large Climate Simulation Datasets," Section on Statistical Computing, JSM2012, San Diego, CA, USA.

[4] **Chen, W.-C.** and Maitra, R. (2011), "A New AECM Algorithm Tuned to A Faster Beat for Model-Based Clustering of Regression Time Series Data," Section on Statistical Learning and Data Mining, JSM2011, Miami, FL, USA. (Student Paper Competition Winner).

[3] **Chen, W.-C.**, Dorman, K.S., Lunney, J.K., and Rowland, R.R.R. (2011), "Model-Based Functional Clustering for Porcine Reproductive and Respiratory Syndrome Virus Data," SCVMMI, Clarke University, Dubuque, IA, USA.

[2] Melnykov, V., **Chen, W.-C.**, and Maitra, R. (2010), "Model-Based Semisupervised Clustering," Biometrics Section, JSM2010, Vancouver, BC, Canada.

[1] **Chen, W.-C.** and Dorman, K.S. (2009), "Twisted Sisters: Disentangling Selection in Overlapping Reading Frames," Section on Statistical Computing, JSM2009, Washington, DC, USA. (Student Paper Competition Winner).

Posters

[8] **Chen, W.-C.**, Li, J., and Scott, J. (2016), "Subgroup Analyses for Count Data using Bayesian Empirical Meta-Analytical Predictive Priors," 2016 ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Washington D.C., USA.

[7] **Chen, W.-C.**, Li, J., and Scott, J. (2016), "Subgroup Analyses for Count Data using Bayesian Empirical Meta-Analytical Predictive Priors," Biopharmaceutical Section, JSM2016, Chicago, IL, USA.

[6] **Chen, W.-C.** and Maitra, R. (2016), "A Novel Model-based Segmentation Approach for Improved Activation Detection in fMRI Studies," ISMRM 24th Annual Meeting, Singapore.

[5] **Chen, W.-C.**, Li, J., Scott, J., and Mintz, P. (2015), "Bayesian-Commensurate Approach for Safety Assessment in Clinical Studies with Count Outcomes," Biopharmaceutical Section, JSM2015, Seattle, WA, USA.

[4] **Chen, W.-C.**, Zaretzki, R., and Gilchrist, M. (2014), "cubfits: an R package for codon usage bias fits," Society for Molecular Biology and Evolution, SMBE2014, San Juan, Puerto Rico.

[3] **Chen, W.-C.**, Ostrouchov, G., Pugmire, D., Prabhat, and Wehner, M. (2012), "Applying model-based clustering for analysis of Community Atmospheric Model output," Conference on Data Analysis (CoDA), Statistical Sciences Group, Los Alamos National Laboratory, Santa Fe, NM, USA.

[2] Dorman, K.S. and **Chen, W.-C.** (2008), "Overlapping Codon Model for Detecting Selection Effects in Overlapping Reading Frames," Fall Conference on Statistics in Biology 2008, Iowa State University, Ames, Iowa, USA.

[1] **Chen, W.-C.**, Lin, C.-Y., Huang, Y.-S., Chang, I.-S., and, Hsiung, C.A. (2004), "Microarray On Line Analysis System (MOLAS): A web platform for array data management and analysis," Intelligent Systems for Molecular Biology, ISMB2004, Glasgow, Scotland, UK.

TEACHING & MENTORING EXPERIENCE

Short Course and Tutorial

[2] **Chen, W.-C.** and Ostrouchov, G. (2017), "Essentials of High Performance and Parallel Statistical Computing with R," Continuing Education Course, JSM2017, Baltimore, MD, USA.

[1] **Chen, W.-C.**, Zaretzki, R., and Gilchrist, M. (2014), "Fitting Complex Mixed Models using HPC," in NIMBioS Tutorial (Computing in the Cloud: What Every Computational Life Scientist Should Know), University of Tennessee, Knoxville, TN, USA.

Mentorship and Student Supervision

- Chunxiao Wang, Stat PhD, Oregon State University. 2017 Summer at FDA/CDRH/OSB/DBS.
- Baosheng He, Industrial Engineering PhD, University of Iowa. 2016 Summer at ORNL/JICS.
- Yuping Lu, EE&CS PhD, University of Tennessee, Knoxville. 2016 Summer at ORNL.
- Christian Heckendorf, CS MS, Boston University. 2014 Summer at Google Summer of Code.
- Gaurav Sehrawat, BS, Jaypee Institute of Information Technology, India. 2013 Summer at Google Summer of Code.
- Cristian Capdevila, CS BS, University of Tennessee, Knoxville. 2012 Summer at ORNL.
- John Aidoo, Applied Stat MS, Wright State University. 2010 Summer at ISU.

PROFESSIONAL SERVICES

Journal Manuscript Reviews

- *Pharmaceutical Statistics* (2017, 2016 ($\times 2$), 2015).
- *Methods in Ecology and Evolution* (2015 ($\times 2$), 2011).
- *Statistics in Biopharmaceutical Research* (2015).
- *Journal of Statistical Software* (2015, 2012).
- *Technometrics* (2013).

Conference Manuscript/Poster Reviews

- *Biopharmaceutical Section Joint Statistical Meeting Poster Competition* (2016 ($\times 2$), 2015 ($\times 4$)).
- *IEEE International Conference on Networking, Architecture, and Storage* (2012).

Conference Organizing Activity

- Session Organizer: "Personalized Medicine: How Bayesian Subgroup Analysis Plays Its Role," 2016 ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Washington D.C., USA.
- Session Chair: "Dimension Reduction and Semiparametric Modeling," IMS, JSM2011, Miami Beach, FL, USA.

HONORS & AWARDS

5 2016 Significant Event Award (SEA) at the Oak Ridge National Laboratory for "Harnessing HPC Capability at OLCF with the R Language for Deep Data Science."

4 2011 Vince Sposito Award for Excellence in Statistical Computing, Department of Statistics, Iowa State University.

3 2011 ASA Statistical Learning and Data Mining Student Paper Competition Award.

2 2010 Preparing Future Faculty, Iowa State University.

1 2009 ASA Statistical Computing/Statistical Graphics Student Paper Competition Award.

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